

Notes: Cabezón & Hoberg (RFS, 2026)

Leaky Director Networks and Innovation Herding

Contents

1	Big picture	3
2	Data and measurement	4
2.1	Board networks and director overlaps	4
2.2	Product-market competitors: TNIC4 and TNIC2-TNIC4	4
2.3	Sample construction	4
2.4	Overlapping-director density	5
2.5	Main dependent variables	5
2.6	Summary statistics and network prevalence	6
3	Models and empirical specifications	6
3.1	Baseline firm-year panel regression	6
3.2	Predicted signs under innovation herding	6
3.3	Distant-peer benchmark	7
3.4	ISS multiboard-director shock	7
3.5	ISS event-study specification	7
3.6	Corporate opportunity waivers	7
3.7	COW parallel-trends decomposition	8
3.8	Technology diffusion regression	8
4	Identification and empirical design	8
4.1	Main identifying variation	8
4.2	Why the ISS shock helps	9
4.3	Why COWs help	9
4.4	Distant competitors and geography as falsification or contrast tests	9
4.5	What the paper cannot fully prove	10
5	Tables and figures	11
5.1	Figure 1: Director overlap and product-market similarity	11
5.2	Table 1 and Table 2: Summary statistics and correlations	11
5.3	Figure 2 and Table 3: Baseline innovation-herding results	12
5.4	Table 4: Distant competitors as a contrast test	13
5.5	Table 5 and Figure 3: ISS multiboard-director shock	14
5.6	Table 6, Table 7, and Figure 4: Corporate opportunity waivers	15
5.7	Table 8: Technology diffusion mechanism	15
6	Short summary	17
7	Conclusion	17
7.1	Core conclusion	17
7.2	Economic intuition	17
7.3	Why the evidence is persuasive	17

7.4 Takeaway	17
------------------------	----

1 Big picture

- **Question.** What happens when directors sit on the boards of direct product-market competitors, and can these overlapping-director networks transmit intellectual property in a way that changes innovation and firm outcomes?
- **Core object.** The paper studies a competitor overlapping-director network, or **CODN**. This is the subnetwork formed by board overlaps among firms that are close competitors according to the 10-K text-based TNIC4 product-market classification.
- **Main mechanism.** Overlapping directors can transmit proprietary information, strategic opportunities, or technical know-how across rival firms. When the network is dense, a technology or business idea can propagate through many competitors rather than remaining proprietary.
- **Central hypothesis.** Dense CODNs generate **innovation herding**: firms become more similar, differentiate less, face more competitive threat, and ultimately perform worse.
- **Important distinction.** Leakage to *close competitors* is harmful because products are substitutes. Leakage to more distant related firms can be beneficial because products may be complements.
- **Baseline evidence.** A higher TNIC4 overlapping-director density predicts higher product similarity, lower ROA, lower market-to-book, higher R&D spending, weaker patent outcomes, and higher product market fluidity.
- **Endogeneity challenge.** Firms may choose directors and innovation strategies jointly. To address this, the paper uses plausibly exogenous shocks to overlapping-director networks and director incentives to leak.
- **ISS shock.** In 2004, ISS changed its voting recommendation policy against directors serving on too many public company boards. This reduced overlapping-director density in affected competitor networks.
- **COW shock.** States adopted corporate opportunity waivers, which weakened legal barriers against directors or officers appropriating business opportunities. Dense CODNs exposed to COWs exhibit stronger innovation-herding effects.
- **Mechanism evidence.** A technology-word diffusion test shows that when a firm's TNIC4 peers use a technology, the focal firm adopts that technology faster if its competitor network has high overlapping-director density. The effect is even stronger when competitor networks are exposed to COWs.
- **Takeaway.** Board interlocks among competitors are not merely governance curiosities. They can create a network through which intellectual property leaks, rivals herd into similar technologies, product differentiation declines, and firm value suffers.

2 Data and measurement

2.1 Board networks and director overlaps

- The paper uses BoardEx and ISS Directors databases to identify directors and board memberships.
- For each year, the authors construct all firm-director pairs and then all firm-pairs that share at least one director.
- Two firms are considered connected by an overlapping director if at least one person sits on both boards in the same year.
- The full economy-wide overlapping-director network is the ODN. The paper’s main focus is the **competitor ODN**, or CODN, which retains only overlaps among direct product-market competitors.

Interpretation: The board network matters because a director with fiduciary knowledge of more than one firm can carry ideas, plans, opportunities, and strategic information across firms. The main empirical question is whether this transfer is especially harmful when the linked firms compete closely.

2.2 Product-market competitors: TNIC4 and TNIC2-TNIC4

- Product-market rivals are measured using Hoberg-Phillips text-based network industry classifications.
- TNIC is built from pairwise cosine similarities of product descriptions in 10-K filings.
- The main competitor set is **TNIC4**. It is calibrated to have the same coarseness as four-digit SIC codes and contains only the top 1% of pairwise product similarities.
- The comparison set of more distant competitors is **TNIC2-TNIC4**: firms in the top 5% of product similarity but not in the top 1%.
- TNIC membership is firm-specific and time-varying. Each firm has a unique set of rivals that can change as product descriptions change.

Interpretation: TNIC4 captures firms whose products are close substitutes. TNIC2-TNIC4 captures related but more distant firms where complementarities are more likely. This distinction is crucial because leakage to substitutes can destroy differentiation, whereas leakage to complements can create value.

2.3 Sample construction

- The main sample covers 1990 to 2019.
- The final sample contains 4,260 firms and 41,300 firm-year observations.
- The paper excludes financial firms and utilities.
- Accounting and market variables come from CRSP/Compustat.
- Patent data come from the Kogan et al. patent database, covering granted USPTO patent applications.
- Variables are winsorized at the 5% and 95% levels.

Interpretation: The sample combines product-market networks, board networks, accounting performance, patent output, and text-based product measures. This allows the paper to trace director-network density into products, innovation, and firm value.

2.4 Overlapping-director density

Let $\mathcal{P}_{i,t}^{TN4}$ denote the set of TNIC4 peers of firm i in year t . Let $N_{i,t} = |\mathcal{P}_{i,t}^{TN4}|$, and let $OD_{jk,t}$ equal one if peer firms j and k share at least one director in year t . The paper’s main density measure can be written as

$$ODdensity_{i,t}^{TN4} = \frac{1}{N_{i,t}(N_{i,t} - 1)/2} \sum_{j < k \in \mathcal{P}_{i,t}^{TN4}} OD_{jk,t}. \quad (1)$$

- The measure is a local clustering coefficient for the focal firm’s competitor network.
- It is the fraction of all peer-pair combinations among the focal firm’s TNIC4 rivals that are linked by overlapping directors.
- A high value means that the focal firm operates in a product market where its direct competitors are heavily connected through board overlaps.
- The same logic is also used to compute density among distant peers, $ODdensity_{i,t}^{TN2-TN4}$.

Interpretation: The variable is not simply whether the focal firm shares a director with someone. It measures the density of the competitor network surrounding the firm. The theory predicts that leakage effects can harm a focal firm even if the focal firm itself is not directly connected, because information can spread among its rivals.

2.5 Main dependent variables

The paper studies four families of outcomes:

- **Product differentiation:**
 - $TSimTN4$: total TNIC4 product similarity, the sum of product similarities between the focal firm and its TNIC4 peers.
 - TNIC HHI in robustness tests: higher concentration indicates greater differentiation; lower concentration indicates more competition.
- **Performance and value:**
 - ROA: operating income before depreciation divided by assets.
 - Market-to-book: market value divided by book value.
- **Innovation inputs and outputs:**
 - R&D divided by assets.
 - Patents divided by assets.
 - Patent value divided by assets.
- **Competitive threat:**
 - Product market fluidity from Hoberg, Phillips, and Prabhala, measuring how much the firm’s product market is changing around it.

Interpretation: The innovation-herding mechanism predicts a distinctive combination: firms may spend more on R&D because leaked ideas are available, but their products become more similar, patents become less valuable, and performance deteriorates.

2.6 Summary statistics and network prevalence

- The mean TNIC4 OD density is about 0.012, meaning that about 1.2% of possible competitor-pair links are direct director overlaps.
- When second-degree connections are considered, the mean density rises to about 0.017.
- For more distant TNIC2-TNIC4 peers, the average density is about 0.007 for first-degree links and 0.013 for second-degree links.
- About 83.9% of firms are connected to the broader OD network through enough degrees of connectedness.
- About 18.3% of sample firms have at least one overlapping director sitting on the board of a TNIC4 direct competitor.

Interpretation: OD density values can look small because the denominator counts all possible peer-pair combinations, which grows quadratically with the number of rivals. In director-appointment terms, the prevalence is economically meaningful.

3 Models and empirical specifications

3.1 Baseline firm-year panel regression

The baseline specification is

$$DEP_{i,t} = \alpha_0 + \beta_1 TN4\ ODdensity_{i,t-1} + \delta' Controls_{i,t-1} + \mu_i + \gamma_t + \epsilon_{i,t}. \quad (2)$$

- $DEP_{i,t}$ is one of the outcome variables: product similarity, ROA, market-to-book, R&D/assets, patents/assets, patent value/assets, or fluidity.
- $TN4\ ODdensity_{i,t-1}$ is standardized and lagged by one year.
- Controls include the logarithm of total assets and the logarithm of firm age. Some specifications also include market-to-book.
- μ_i are firm fixed effects.
- γ_t are year fixed effects.
- Standard errors are double clustered by industry-year and by firm.

Interpretation: Firm fixed effects remove time-invariant firm heterogeneity, while year fixed effects remove common shocks. The coefficient β_1 is identified from within-firm changes in the density of the firm's competitor-director network.

3.2 Predicted signs under innovation herding

For direct competitors, the innovation-herding hypothesis predicts:

$$\beta_1(TSimTN4) > 0, \quad (3)$$

$$\beta_1(ROA) < 0, \quad (4)$$

$$\beta_1(Market-to-Book) < 0, \quad (5)$$

$$\beta_1(R\&D/Assets) \geq 0, \quad (6)$$

$$\beta_1(Patent\ value/Assets) < 0, \quad (7)$$

$$\beta_1(Fluidity) > 0. \quad (8)$$

Interpretation: The mechanism does not predict a simple decline in all innovative activity. Instead, firms can spend more on R&D to exploit leaked ideas, but because many rivals receive similar ideas, differentiation and valuable patent output decline.

3.3 Distant-peer benchmark

The paper also estimates a specification with both close-competitor and distant-peer network density:

$$DEP_{i,t} = \alpha_0 + \beta_1 TN4\ ODdensity_{i,t-1} + \beta_2 TN2-TN4\ ODdensity_{i,t-1} + \delta' Controls_{i,t-1} + \mu_i + \gamma_t + \epsilon_{i,t}. \quad (9)$$

Interpretation: Distant rivals are more likely to be complements rather than substitutes. If leakage to complements is value creating, β_2 should have the opposite sign of β_1 for many outcomes.

3.4 ISS multiboard-director shock

In 2004, ISS began recommending WITHHOLD votes against directors serving on more than six public company boards, and against CEO-directors serving on more than three including their own firm. The paper uses this as a shock to highly connected directors.

The first-stage regression is

$$TN4\ ODdensity_{i,t} = \beta_1 (TN4\ ISS\ Treated_i \times Post2004_t) + \delta' Controls_{i,t-1} + \mu_i + \gamma_t + \epsilon_{i,t}. \quad (10)$$

The second-stage regression is

$$DEP_{i,t+1} = \beta_1 TN4\ \widehat{ODdensity}_{i,t} + \mu_i + \gamma_t + \epsilon_{i,t}. \quad (11)$$

- Treatment is defined at the focal firm's TNIC4 network, not only at the focal firm itself.
- The sample window is 1999–2008, five years before and after the shock.
- The design is an intent-to-treat design: treatment is based on exposure to the rule regardless of actual board changes.

Interpretation: This design helps address endogeneity in board-network formation. The ISS rule reduces network density for plausibly external reasons, allowing the paper to study how lower OD density changes product and innovation outcomes.

3.5 ISS event-study specification

The paper also estimates event-time effects around the ISS shock:

$$TN4\ ODdensity_{i,t} = \beta_{1999} (TN4\ ISS\ Treated_i \times Year1999_t) + \dots \quad (12)$$

$$+ \beta_{2007} (TN4\ ISS\ Treated_i \times Year2007_t) + \delta' Controls_{i,t-1} + \mu_i + \gamma_t + \epsilon_{i,t}. \quad (13)$$

Interpretation: The purpose is to test whether treated and untreated networks were on parallel trends before 2004 and whether the decline in OD density occurs after the ISS rule change.

3.6 Corporate opportunity waivers

The corporate opportunity doctrine limits the ability of directors and officers to appropriate business opportunities belonging to the corporation. Some states adopted corporate opportunity waivers (COWs), allowing firms to waive these protections in advance.

The paper estimates

$$DEP_{i,t} = \alpha_0 + \beta_1 TN4\ ODdensity_{i,t-1} + \beta_2 (TN4\ ODdensity_{i,t-1} \times TN4\ COW_{i,t-1}) \quad (14)$$

$$+ \beta_3 TN4\ COW_{i,t-1} + \delta' Controls_{i,t-1} + \mu_i + \gamma_t + \epsilon_{i,t}. \quad (15)$$

- $TN4\ COW_{i,t-1}$ equals one when the percentage of the focal firm's TNIC4 rivals affected by COW is above the median.
- The key coefficient is β_2 .
- The prediction is that director-network density should have stronger negative consequences when directors have greater legal scope or incentive to appropriate opportunities.

Interpretation: COWs do not simply reduce competition directly. In the paper's mechanism, they make leakage through directors more consequential by weakening legal barriers to taking or redirecting corporate opportunities.

3.7 COW parallel-trends decomposition

To test parallel trends around staggered COW adoption, the paper decomposes OD density into event-time components:

$$DEP_{i,t} = \alpha_0 + \beta_0 TN4\ ODDensity[Never\ Treated]_{i,t} \quad (16)$$

$$+ \beta_1 TN4\ ODDensity[ODs\ treated\ 20\ years\ from\ now]_{i,t} + \dots \quad (17)$$

$$+ \beta_{20} TN4\ ODDensity[Treated\ in\ T]_{i,t} + \dots \quad (18)$$

$$+ \beta_{41} TN4\ ODDensity[ODs\ treated\ 20\ years\ ago]_{i,t} + \delta' Controls_{i,t-1} + \mu_i + \gamma_t + \epsilon_{i,t}. \quad (19)$$

Interpretation: The parallel-trends test asks whether the coefficients are stable before COW treatment and change after treatment. This is a network-level event-study design, because treatment is tied to the event timing of firms connected to the focal firm's network.

3.8 Technology diffusion regression

For a focal firm i , technology k , and year t , the paper constructs a dummy for whether firm i mentions technology k in its 10-K. It then estimates specifications of the form

$$AdoptTech_{i,k,t} = \alpha_i + \lambda_{k,t} + \beta_1 PeerTech_{i,k,t-1} + \beta_2 ODDensity_{i,t-1} \quad (20)$$

$$+ \beta_3 (PeerTech_{i,k,t-1} \times ODDensity_{i,t-1}) + \beta_4 COW_{i,t-1} \quad (21)$$

$$+ \beta_5 (PeerTech_{i,k,t-1} \times COW_{i,t-1}) \quad (22)$$

$$+ \beta_6 (PeerTech_{i,k,t-1} \times ODDensity_{i,t-1} \times COW_{i,t-1}) + \epsilon_{i,k,t}. \quad (23)$$

- $PeerTech_{i,k,t-1}$ is the percentage of TNIC4 peers using technology k in the previous year.
- $ODDensity_{i,t-1}$ is the focal firm's TNIC4 OD density.
- $COW_{i,t-1}$ is COW exposure among TNIC4 peers.
- The regressions include firm fixed effects and technology-by-year fixed effects.

Interpretation: This is the most direct mechanism test. If technology adoption propagates faster through dense CODNs, then peer technology adoption should predict focal firm adoption more strongly when OD density is high, and even more strongly when COW exposure is high.

4 Identification and empirical design

4.1 Main identifying variation

- The baseline design uses within-firm changes in the density of the competitor OD network.

- Firm fixed effects absorb permanent firm-level features such as business model, industry position, or governance style.
- Year fixed effects absorb aggregate shocks to innovation, boards, and product markets.
- Lagged OD density reduces simultaneity between contemporaneous outcomes and network structure.

Interpretation: The baseline regressions establish a robust association between dense CODNs and innovation herding, but the paper does not rely on them alone for causal interpretation.

4.2 Why the ISS shock helps

- The ISS rule targeted directors with many board seats, not a specific firm’s innovation choices.
- Treated networks are defined using TNIC4 peers, so treatment is less directly tied to the focal firm’s own actions.
- The first stage shows that the rule reduced TNIC4 OD density.
- The second stage asks whether the predicted change in OD density produces the same pattern as the baseline regressions.

Interpretation: The ISS design turns a governance rule into variation in director-network density. It supports the interpretation that OD density itself contributes to the outcomes rather than merely reflecting them.

4.3 Why COWs help

- Corporate opportunity waivers change directors’ legal ability or incentive to redirect opportunities.
- COW adoption is staggered across states.
- The paper measures network exposure using the share of TNIC4 peers affected by COWs.
- If leakage drives the results, OD density should become more harmful when networks are more exposed to COWs.

Interpretation: The COW design tests a mechanism, not just an outcome. If the effect of OD density intensifies when directors face weaker legal constraints on opportunity diversion, the leakage mechanism becomes more plausible.

4.4 Distant competitors and geography as falsification or contrast tests

- Distant TNIC2-TNIC4 peers provide a product-market contrast: they are related but less likely to be direct substitutes.
- Geography tests ask whether results are driven by local knowledge spillovers rather than board-network leakage.
- The paper shows that geographically separated OD-density measures still produce the core results.
- A city-density measure does not replicate the negative CODN results and in some cases has the opposite sign.

Interpretation: These tests are important because knowledge spillovers can be good. The paper argues that the harmful effect is specifically about leakage to close competitors, not knowledge sharing in general.

4.5 What the paper cannot fully prove

- Director overlap is not randomly assigned in the baseline regressions.
- The paper observes technology diffusion through 10-K language, not private boardroom conversations.
- Product similarity can rise for reasons unrelated to leakage, although the ISS and COW designs mitigate this concern.
- The legal interpretation of COWs and corporate opportunity doctrine is institutionally rich and may vary across firms.
- The results are strongest for public firms with available board, product-text, and patent data.

Interpretation: The paper's strongest evidence comes from triangulation: baseline panel regressions, distant-peer contrasts, ISS shocks, COW shocks, parallel-trends figures, and direct technology diffusion tests all point toward the same mechanism.

5 Tables and figures

This section follows the *Read* plus *Economic message* format. All visuals are directly cropped from the paper and arranged compactly.

5.1 Figure 1: Director overlap and product-market similarity

Read:

- Figure 1 shows the incidence of overlapping directors as a function of product-market similarity.
- The incidence of OD links rises as firms become more similar in product space.
- The effect is especially striking among the most similar firm-pairs, which are likely to be direct competitors.
- The figure documents that the practice exists despite legal and ethical concerns around competing-firm director overlaps.

Economic message: Firms do not avoid board overlaps with direct competitors. This creates the empirical setting in which proprietary information could leak through competitor-board networks.

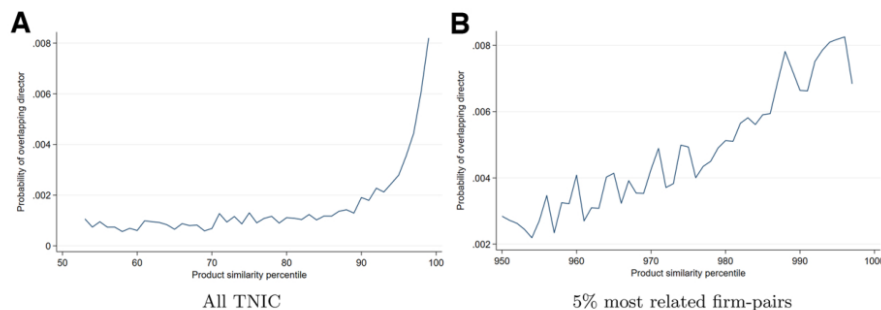


Figure 1

Director overlap by percentile of TNIC similarity

The figure shows the average director overlap for firm-pairs that are in the same percentile of TNIC product similarity. Panel A uses all firms. Panel B focuses on pair of firms with similarity above the 95th percentile.

5.2 Table 1 and Table 2: Summary statistics and correlations

Read:

- Table 1 reports the main sample and variable distributions.
- Mean TNIC4 OD density is about 0.012, while second-degree OD density is higher.
- Distant-peer OD density is lower on average than close-competitor OD density.
- Table 2 gives the correlation matrix for OD density, product similarity, performance, innovation, and competition measures.

Economic message: The summary statistics show that competitor board overlaps are prevalent enough to matter, while the correlations motivate the formal panel regressions.

Table 1
Summary statistics

	Obs	Mean	Std. dev.	Min	Max	P50
1st-degree OD density TNIC4	41,337	.012	.012	0	0.052	0.010
1st-degree OD density TNIC2-TNIC4	41,337	.007	.003	0	.024	.007
2nd-degree OD density TNIC4	41,337	.017	.02	0	.08	.010
1st-degree OD focal TNIC4	41,337	.007	.018	0	.083	0
1 if connected to ODN	41,337	.839	.367	0	1	1
1 if connected to CODN	41,337	.183	.387	0	1	0
XRD/(total assets)	41,337	.046	.071	0	.378	.005
Number of patents/(total assets)	41,337	.007	.015	0	.114	0
Total value patents/(total assets)	41,337	.032	.083	0	.765	0
TNIC4 total product similarity	41,337	2.606	5.266	.008	45.654	.689
Product market fluidity	41,337	6.686	3.213	2.002	20.274	6.054
ROA	41,337	.093	.134	-.583	.311	.117
Market-to-book ratio	41,337	1.702	1.153	.159	6.946	1.318
log of total assets	41,337	6.514	1.806	3.041	10.702	6.477
log of age	41,337	2.509	.875	.693	4.007	2.639

This table presents the summary statistics of the main variables. Financial firms and utilities are excluded. Total assets, R&D expenditure, firm age, market-to-book ratio, and ROA are financial indicators from Compustat. Firm age is computed using the first effective date of the current link. R&D was set to zero when missing. ROA is the operating income before depreciation (oibdp) over assets. All nonbinary variables are yearly winsorized at the 5%-95% level.

(a) Table 1: summary statistics.

Table 2
Correlations

	TN4 OD density	TN4 2deg OD density	TN4 ODfocal	XRD/at	pat/at	val pat/at	TSim TN4	Fluidity	ROA	mkt/book
TN4 OD density	1.000									
TN4 2deg OD density	0.406	1.000								
TN4 ODfocal	0.346	0.172	1.000							
XRD/at	0.255	0.227	0.164	1.000						
Pat/at	0.140	0.051	0.096	0.465	1.000					
Val pat/at	0.107	0.050	0.099	0.324	0.664	1.000				
TSimTN4	0.245	0.317	0.215	0.457	0.135	0.125	1.000			
Fluidity	0.224	0.190	0.188	0.347	0.163	0.127	0.627	1.000		
ROA	-0.164	-0.192	-0.107	-0.531	-0.166	0.014	-0.394	-0.310	1.000	
Mkt/book	0.111	0.084	0.083	0.355	0.236	0.328	0.224	0.179	0.040	1.000

This table displays the correlation matrix of the main variables.

(b) Table 2: correlation matrix.

5.3 Figure 2 and Table 3: Baseline innovation-herding results

Read:

- Figure 2 shows that TNIC4 director overlap increased in the early 1990s, was relatively stable in the 2000s, and rose further after 2010.
- Table 3 panel A shows that TNIC4 OD density predicts higher total product similarity, lower ROA, lower market-to-book, higher R&D/assets, and higher fluidity.
- The coefficient on TNIC4 OD density is 0.012 for total similarity, -0.008 for ROA, -0.010 for market-to-book, 0.004 for R&D/assets, and 0.022 for fluidity.
- Patent counts are not significantly increased, and patent value is weak or negative, consistent with redundant innovation rather than valuable new inventions.
- Panels B–D show that the results are robust to stricter definitions of whether a firm is connected to the

OD network or to alternative employment-based director links.

Economic message: Dense CODNs make competitors more similar, less profitable, less valuable, and more exposed to competitive threat. This is the baseline evidence for innovation herding.

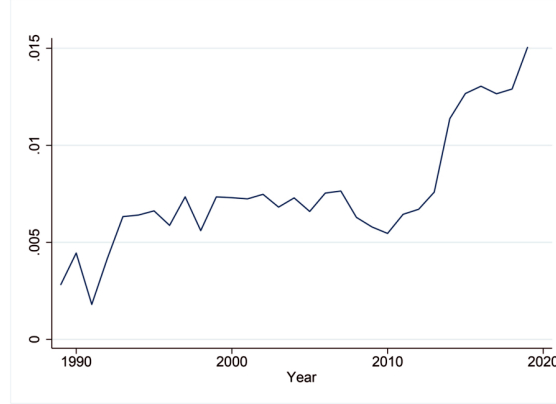


Figure 2
TNIC4 director overlap over time
The figure shows a time-series plot of the average director overlap for TNIC4 firm-pairs.

Table 3
TNIC4 OD density and firm outcomes
A. Baseline

	(1) TSim TN4 _t	(2) ROA _t	(3) Mkt/ Book _t	(4) XRD _t / at _t	(5) Pat _t / at _t	(6) ValPat _t / at _t	(7) Fluid _t
TN4 OD density in t - 1	0.012*** (0.003)	-0.008** (0.004)	-0.010** (0.005)	0.004* (0.002)	0.003 (0.005)	-0.010 (0.006)	0.022*** (0.004)
log of total assets in t - 1	0.049*** (0.011)	-0.007 (0.017)	-0.252*** (0.015)	-0.110*** (0.010)	-0.086*** (0.018)	-0.123*** (0.023)	0.079*** (0.012)
log of firm age in t - 1	-0.045*** (0.015)	-0.001 (0.017)	-0.076*** (0.021)	0.028** (0.011)	-0.141*** (0.021)	-0.005 (0.027)	-0.165*** (0.019)
Observations	41,302	42,272	42,122	42,323	42,323	42,323	41,774
R-squared	.863	.716	.656	.906	.654	.614	.800
Adj within R-squared	.006	.000	.041	.024	.011	.007	.016
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes

B. TNIC4 OD density is set to zero if the firm is not connected to ODN

	(1) TSim TN4 _t	(2) ROA _t	(3) Mkt/ Book _t	(4) XRD _t / at _t	(5) Pat _t / at _t	(6) ValPat _t / at _t	(7) Fluid _t
TN4 OD density in t - 1	0.012*** (0.003)	-0.009** (0.004)	-0.011** (0.005)	0.003 (0.003)	-0.004 (0.005)	-0.019*** (0.007)	0.020*** (0.005)
log of total assets in t - 1	0.049*** (0.011)	-0.007 (0.017)	-0.251*** (0.015)	-0.110*** (0.010)	-0.085*** (0.018)	-0.122*** (0.023)	0.078*** (0.012)
log of firm age in t - 1	-0.045*** (0.015)	-0.001 (0.017)	-0.076*** (0.021)	0.028** (0.011)	-0.141*** (0.021)	-0.004 (0.027)	-0.166*** (0.019)
Observations	41,302	42,272	42,122	42,323	42,323	42,323	41,774
R-squared	.863	.716	.656	.906	.654	.614	.800
Adj within R-squared	.006	.000	.041	.024	.011	.008	.016
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes

(Continued)

Table 3
(Continued)

C. TNIC4 OD density is set to zero if the firm is not connected to CODN

	(1) TSim TN4 _t	(2) ROA _t	(3) Mkt/ Book _t	(4) XRD _t / at _t	(5) Pat _t / at _t	(6) ValPat _t / at _t	(7) Fluid _t
TN4 OD density in t - 1	0.038*** (0.006)	-0.021*** (0.007)	-0.007 (0.005)	0.004 (0.005)	-0.012 (0.009)	-0.040*** (0.012)	0.034*** (0.006)
log of total assets in t - 1	0.047*** (0.010)	-0.006 (0.017)	-0.252*** (0.015)	-0.110*** (0.010)	-0.085*** (0.018)	-0.120*** (0.023)	0.077*** (0.012)
log of firm age in t - 1	-0.046*** (0.015)	-0.000 (0.017)	-0.076*** (0.021)	0.028** (0.011)	-0.140*** (0.021)	-0.003 (0.027)	-0.167*** (0.019)
Observations	41,302	42,272	42,122	42,323	42,323	42,323	41,774
R-squared	.863	.716	.656	.906	.654	.615	.800
Adj within R-squared	.011	.001	.040	.024	.011	.009	.018
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes

D. TNIC4 OD density is set to zero if the firm is not connected to ODN (through OD or other associations)

	(1) TSim TN4 _t	(2) ROA _t	(3) Mkt/ Book _t	(4) XRD _t / at _t	(5) Pat _t / at _t	(6) ValPat _t / at _t	(7) Fluid _t
TN4 OD density in t - 1	0.011*** (0.003)	-0.011** (0.004)	-0.012** (0.005)	0.004* (0.003)	-0.000 (0.005)	-0.015** (0.007)	0.020*** (0.005)
log of total assets in t - 1	0.049*** (0.011)	-0.007 (0.017)	-0.251*** (0.015)	-0.110*** (0.010)	-0.086*** (0.018)	-0.122*** (0.023)	0.078*** (0.012)
log of firm age in t - 1	-0.045*** (0.015)	-0.001 (0.017)	-0.076*** (0.021)	0.028** (0.011)	-0.141*** (0.021)	-0.004 (0.027)	-0.166*** (0.019)
Observations	41,302	42,272	42,122	42,323	42,323	42,323	41,774
R-squared	.863	.716	.656	.906	.654	.614	.800
Adj within R-squared	.006	.000	.041	.024	.011	.008	.016
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes

5.4 Table 4: Distant competitors as a contrast test

Read:

- Table 4 adds OD density among more distant TNIC2-TNIC4 peers.
- Distant-peer OD density has signs almost opposite to direct-competitor OD density.
- TNIC2-TNIC4 OD density is associated with lower total similarity, higher ROA, higher market-to-book, higher patent value, and lower fluidity.
- Direct TNIC4 OD density remains harmful in the same regressions.

Economic message: The result shows that director-network information transfer is not always bad. It is harmful when firms are close substitutes, but potentially beneficial when firms are more distant complements.

Table 4
Outer ring (TNIC2-TNIC4) OD density and firm outcomes

	(1) TSim TN4 _t	(2) ROA _t	(3) Mkt/ Book _t	(4) XRD _t / at _t	(5) Pat _t / at _t	(6) ValPat _t / at _t	(7) Fluid _t
TN2-TN4 OD density in $t-1$	-0.023*** (0.004)	0.013*** (0.005)	0.011** (0.006)	0.002 (0.003)	0.005 (0.006)	0.029*** (0.007)	-0.037*** (0.005)
TN4 OD density in $t-1$	0.013*** (0.003)	-0.008** (0.004)	-0.011** (0.005)	0.004* (0.002)	0.003 (0.005)	-0.010* (0.006)	0.023*** (0.004)
log of total assets in $t-1$	0.049*** (0.010)	-0.007 (0.017)	-0.251*** (0.015)	-0.110*** (0.010)	-0.085*** (0.018)	-0.123*** (0.023)	0.078*** (0.012)
log of firm age in $t-1$	-0.044*** (0.015)	-0.002 (0.017)	-0.077*** (0.021)	0.028** (0.011)	-0.142*** (0.021)	-0.007 (0.027)	-0.165*** (0.019)
Observations	41,266	42,213	42,063	42,264	42,264	42,264	41,720
R-squared	.863	.716	.657	.906	.654	.614	.801
Adj within R-squared	.908	.900	.941	.924	.911	.908	.919
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes

This table presents the results of the OLS panel regression of different firm policies on TNIC4 OD density (standardized) and TNIC2-TNIC4 OD density (standardized). In column 1, the dependent variable is the firm's total product similarity in its TNIC4 group. In columns 2 and 3, the dependent variables are ROA and market-to-book ratio. In column 4, it is the firm's XRD/(total assets). In column 5, it is the number of patents divided by total assets. In column 6, it is the total value of patents divided by total assets. In column 7, it is product-market fluidity from [Hoberg, Phillips, and Prabhala \(2014\)](#). Firm and year fixed effects are included. Controls include the logarithm of assets in $t-1$, logarithm of firm age in $t-1$, and market-to-book ratio in $t-1$. All variables are standardized (z -score) and winsorized at the 5%-95% level. Standard errors are double clustered by Industry $\times$ Year and by Firm, and reported in parentheses. * $p < .1$; ** $p < .05$; *** $p < .01$.

5.5 Table 5 and Figure 3: ISS multiboard-director shock

Read:

- Table 5 panel A shows the first stage: the 2004 ISS multiboard-director rule lowers TNIC4 OD density by -0.072 .
- Instrument diagnostics are supportive: the Cragg-Donald F-statistic is 27.70 and the Kleibergen-Paap F-statistic is 11.20.
- Panel B shows that predicted OD density produces the same core pattern as the baseline: higher similarity, lower ROA, lower market-to-book, lower patent value, and higher fluidity.
- Figure 3 presents event-time evidence around the ISS shock and supports the timing of the treatment effect.

Economic message: The ISS rule provides quasi-experimental variation in network density. The results strengthen the interpretation that OD density causes innovation herding and weaker firm outcomes.

Table 5
2004 ISS rule and TNIC4 OD density
A. First stage

	(1) TN4 OD density _t
(TN4 ISS treated)x(post2004)	-0.072*** (0.019)
log of total assets in t - 1	0.081*** (0.019)
log of firm age in t - 1	-0.050 (0.040)
Observations	18,676
R-squared	.518
Adj within R-squared	.004
Firm FE	Yes
Year FE	Yes
Cragg-Donald Wald F-statistic	27.70
Kleibergen-Paap Wald rk F-statistic	11.20
Kleibergen-Paap rk LM statistic	31.03
Anderson-Rubin Wald test	26.85
Stock-Wright LM S statistic	23.99

B. Second stage

	(1) TSim TN4 _t	(2) ROA _t	(3) Mkt/ Book _t	(4) XRD _t / at _t	(5) Pat _t / at _t	(6) ValPat _t / at _t	(7) Fluid _t
TN4 OD density in t - 1 (predicted)	0.171** (0.078)	-0.092*** (0.026)	-1.778*** (0.361)	-0.368*** (0.123)	-0.118 (0.165)	-0.234* (0.124)	0.430*** (0.149)
Observations	18,676	18,676	18,676	18,676	18,676	18,676	18,676
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes

This table presents the results of a 2SLS estimation using the 2004 new ISS rule against multiboard directors. Panel A shows the first stage and panel B the second stage. In the first stage, we estimate a difference-in-difference regression using our TN4 OD density as the dependent variable, a dummy variable that equals one if the focal firm's network is affected by the ISS rule in 2003 as the treatment, and post 2004 as the posttreatment period. In the second stage, we regress our main dependent variables on the OD density predicted values from the first stage. Firm and year fixed effects are included. Controls include the logarithm of assets in t - 1 and logarithm of firm age in t - 1. All variables are standardized (z-score) and winsorized at the 5%-95% level. Standard errors are double clustered by Industry x Year and by Firm, and reported in parentheses. * $p < .1$; ** $p < .05$; *** $p < .01$.

(a) Table 5: ISS shock and 2SLS.

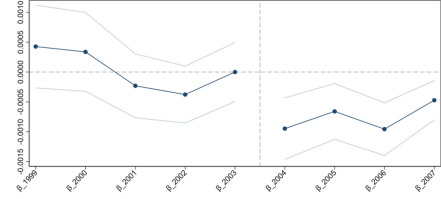


Figure 3
TNIC4 OD density and ISS rule: Parallel trends
The figure illustrates parallel trends analysis around the multiboard directors ISS rule adopted in 2004. The blue line reports the estimated β_t coefficients of Equation (4). The estimation includes firm and year fixed effects, and cluster standard errors by Industry x Year and Firm. The gray line reports the 95% confidence interval.

(b) Figure 3: ISS event study.

5.6 Table 6, Table 7, and Figure 4: Corporate opportunity waivers

Read:

- Table 6 lists the staggered adoption of corporate opportunity waivers: Delaware in 2000, Oklahoma in 2001, Missouri in 2003, Kansas in 2005, Texas in 2006, Nevada in 2007, New Jersey in 2011, Maryland in 2014, and Washington in 2016.
- Table 7 interacts OD density with COW exposure among TNIC4 peers.
- The OD density $\times$ COW interaction is positive for product similarity, R&D, and fluidity, and negative for ROA, market-to-book, patents, and patent value.
- For example, the interaction is 0.028 for total similarity, -0.017 for ROA, -0.028 for market-to-book, -0.062 for patents/assets, -0.109 for patent value/assets, and 0.032 for fluidity.
- Figure 4 provides parallel-trends evidence around COW treatment years.

Economic message: When directors have greater legal scope to appropriate corporate opportunities, dense competitor director networks become even more damaging. This directly supports the leakage mechanism.

5.7 Table 8: Technology diffusion mechanism

Read:

- Table 8 uses a firm-technology-year panel based on 699 technology phrases extracted from 10-Ks.
- The dependent variable equals one if the focal firm uses a given technology in year t .

Table 6
Corporate opportunities waiver adoption timeline

State	Date
Delaware	July 1, 2000
Oklahoma	November 1, 2001
Missouri	October 1, 2003
Kansas	January 1, 2005
Texas	January 1, 2006
Nevada	October 1, 2007
New Jersey	March 11, 2011
Maryland	October 1, 2014
Washington	January 1, 2016

This table presents the states that adopted COW and the date of the adoption.

Table 7
TNIC4 OD density and corporate opportunities waivers

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	TSim TN4 _{<i>t</i>}	ROA _{<i>t</i>}	Mkt/ Book _{<i>t</i>}	XRD _{<i>t</i>} / at _{<i>t</i>}	Pat _{<i>t</i>} / at _{<i>t</i>}	ValPat _{<i>t</i>} / at _{<i>t</i>}	Fluid _{<i>t</i>}
(TN4 OD density in $t-1$)	0.028***	-0.017**	-0.028***	0.015***	-0.063***	-0.109***	0.032***
x(TN4 COW in $t-1$)	(0.006)	(0.008)	(0.010)	(0.005)	(0.013)	(0.018)	(0.007)
TN4 OD density in $t-1$	0.002	-0.002	0.000	-0.002	0.025***	0.030***	0.010**
	(0.003)	(0.005)	(0.006)	(0.003)	(0.007)	(0.009)	(0.005)
TN4 COW in $t-1$	0.037***	-0.012	-0.026**	0.001	-0.021	-0.091***	0.054***
	(0.007)	(0.010)	(0.013)	(0.006)	(0.014)	(0.019)	(0.010)
log of total assets in $t-1$	0.047***	-0.006	-0.250***	-0.110***	-0.083***	-0.118***	0.076***
	(0.010)	(0.017)	(0.015)	(0.010)	(0.018)	(0.023)	(0.012)
log of firm age in $t-1$	-0.046***	-0.001	-0.075***	0.02***	-0.139***	-0.001	-0.169***
	(0.015)	(0.017)	(0.021)	(0.011)	(0.021)	(0.027)	(0.019)
Observations	41,226	42,189	42,041	42,240	42,240	42,240	41,690
<i>R</i> -squared	.863	.716	.657	.906	.655	.617	.801
Adj within <i>R</i> -squared	.008	.000	.041	.025	.013	.014	.019
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes

This table presents the results of the OLS panel regression of different firm policies on TNIC4 OD density. The estimation includes an interaction term between TN4 OD density and a dummy that equals one if the percentage of TNIC4 rivals affected by Corporate Opportunities Waiver is above the median (TN4 COW). In column 1, the dependent variable is the firm's total product similarity in its TNIC4 group. In columns 2 and 3, the dependent variables are ROA and market-to-book ratio. In column 4, it is the firm's XRD/(total assets). In column 5, it is the number of patents divided by total assets. In column 6, it is the total value of patents divided by total assets. In column 7, it is product-market fluidity from [Hoberg, Phillips, and Prabhala \(2014\)](#). Firm and year fixed effects are included. Controls include the logarithm of assets in $t-1$, logarithm of firm age in $t-1$, and market-to-book ratio in $t-1$. All variables are standardized (z -score) and winsorized at the 5%-95% level. Standard errors are double clustered by Industry $\times$ Year and by Firm, and reported in parentheses. * $p < .1$; ** $p < .05$; *** $p < .01$.

- If TNIC4 peers used the technology in $t-1$, the focal firm is more likely to adopt it in t .
- The peer technology coefficient is about 0.557–0.564 across columns.
- The interaction between peer technology use and OD density is positive and significant in columns 1 and 3, around 0.060–0.064.
- The triple interaction with OD density and COW exposure is 0.136 and significant in column 4.

Economic message: This is the cleanest mechanism test. Technologies propagate faster through dense CODNs, and even faster when COW exposure increases director incentives to leak.

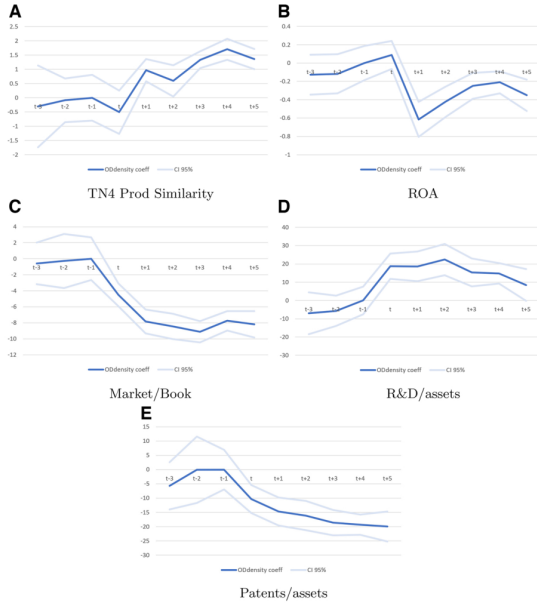


Figure 4
TNIC4 OD density and corporate opportunities waivers: Parallel trends
The figure illustrates parallel trends analysis using a 7-year event study window around COW adoptions. Each figure reports the coefficient of highly targeted OD density variables, where OD density is separately calculated only using OD links of firms treated by COW in a certain event-year. The left-most data point in each graph is based on OD links for firms that will be treated three years into the future. The remaining six data points are based on OD links treated 2 years into the future ("t-2") through the year of treatment ("t"), and through the third year after treatment ("t+3"). All of these coefficients are estimated in a regression that includes separate terms for each of the OD links in each year starting from 20 years pretreatment to 20 years posttreatment. This ensures that each coefficient only compares treated firms in each year to those OD links that are untreated. In particular, the regression also includes the OD density of firms that are never treated by COW. Additionally, the regressions include all the controls included in [Equation \(1\)](#), firm fixed effects, and year fixed effects.

Table 8
TNIC4 OD density and technology diffusion

	(1)	(2)	(3)	(4)
	1 if focal firm adopts technology in t			
TN4 % peers have tech in $t-1$	0.557***	0.564***	0.543***	0.561***
	(0.014)	(0.015)	(0.016)	(0.017)
TN4 OD density in $t-1$	-0.001		-0.000	-0.001
	(0.002)		(0.002)	(0.002)
(Peers have tech in $t-1$)	0.064***		0.060***	0.010
	(0.017)		(0.017)	(0.019)
x(OD density in $t-1$)		0.000	0.000	-0.001
TN4 COW in $t-1$		(0.002)	(0.002)	(0.002)
(Peers have tech in $t-1$)		0.054***	0.051***	-0.003
		(0.016)	(0.015)	(0.018)
x(TN4 COW in $t-1$)				0.005**
				(0.003)
(Peers have tech in $t-1$)				0.136***
				(0.026)
x(TN4 OD density in $t-1$)x(TN4 COW in $t-1$)				
Observations	835,576	777,998	777,998	777,998
<i>R</i> -squared	.205	.205	.205	.205
Adj within <i>R</i> -squared	.087	.087	.088	.088
Firm FE	Yes	Yes	Yes	Yes
Technology $\times$ Year FE	Yes	Yes	Yes	Yes

This table presents the results of an OLS panel regression. The dependent variable is a dummy that equals one if the focal firm has a specific technology in year t . All RHS variables are from year $t-1$ and are characteristics of the TNIC4 network around the focal firm. One observation is a firm-technology-year. We include three key RHS variables in addition to firm fixed effects, year fixed effects and technology fixed effects. The first key variable is the percentage of the focal firm's TNIC4 rivals that have the given technology in $t-1$. The second is the TNIC4 OD density as used in earlier tables. The third is the fraction of the firm's TNIC4 rivals that have been treated by the Corporate Opportunities Waiver (COW) shock based on their state of incorporation and the year $t-1$. We also include double and triple interactions with these variables. All variables are winsorized at the 5%-95% level. Standard errors are double clustered by Industry $\times$ Year and by Firm, and reported in parentheses. * $p < .1$; ** $p < .05$; *** $p < .01$.

6 Short summary

- The paper studies whether overlapping directors among direct competitors create innovation herding.
- It builds competitor overlapping-director networks using BoardEx/ISS and TNIC4 product-market classifications.
- TNIC4 OD density measures how densely a focal firm’s direct competitors are connected through overlapping directors.
- Dense CODNs predict higher product similarity, lower ROA, lower market-to-book, higher R&D, weaker patent outcomes, and higher product market fluidity.
- Distant-peer OD density has mostly opposite effects, suggesting that information sharing can be beneficial when firms are complements rather than close substitutes.
- The 2004 ISS rule against multiboard directors provides a shock that reduces OD density and supports a causal interpretation.
- Corporate opportunity waivers amplify the negative consequences of OD density, supporting the leakage mechanism.
- Technology-diffusion tests show that technologies propagate faster through dense CODNs, especially under COW exposure.
- The overall message is that director networks can spread proprietary technologies, reduce differentiation, and lower firm performance.

7 Conclusion

7.1 Core conclusion

Cabazon and Hoberg show that overlapping directors among competitors can create a leaky network that transmits proprietary information and intellectual property. Dense competitor director networks are associated with innovation herding, lower product differentiation, lower profitability, weaker valuations, and more competitive product markets.

7.2 Economic intuition

A director who sits on multiple boards can transmit ideas, opportunities, and know-how across firms. When the linked firms are close product-market substitutes, this transfer erodes the recipient firms’ and source firms’ ability to differentiate. Many firms may converge on similar innovations, which increases R&D effort but reduces the novelty and proprietary value of innovation.

7.3 Why the evidence is persuasive

The paper combines several layers of evidence: baseline firm-year regressions, a distant-peer contrast, the 2004 ISS shock, COW interactions, COW event-study evidence, geography checks, and a direct technology-diffusion test using 10-K text. The results point in the same direction: the harmful effect is specifically tied to dense director networks among direct competitors.

7.4 Takeaway

The paper reframes board interlocks as a product-market information network. Director connections can be useful when firms are complementary, but they can be damaging when they connect direct competitors. Corporate governance, antitrust, and innovation policy therefore need to consider not just whether a director is conflicted at one firm, but how dense the broader competitor director network is.